

CONTACT
INFORMATION

Department of Computer Science and Operations Research
University of Montréal

Mila, Quebec AI Institute
6666 St. Urbain, Montréal

E-mail: ioannis@iro.umontreal.ca
Web: mitliagkas.github.io

RESEARCH
INTERESTS

Statistical machine learning, optimization, deep learning, game theory, generative models, distributed learning systems.

AFFILIATIONS

University of Montréal, Department of Computer Science and Operations Research

Associate professor

June 2022 -

Assistant professor

Sept. 2017 - May 2022

Mila, Quebec AI Institute

September 2017 -

Core member, Canada CIFAR AI chair

Member of Mila's Scientific Council

Google DeepMind, Montréal, Staff Research Scientist (part time) September 2022-

Archimedes, AI Institute

January 2024-

Affiliate researcher

PAST AFFILIATIONS

ElementAI, Montréal, Faculty Fellow (part time)

2018-2020

Stanford University, Postdoc, Departments of CS and Statistics

2015-2017

Supervised by: Associate prof. Christopher Ré, Adjunct prof. Lester Mackey

EDUCATION

The University of Texas at Austin

PhD, ECE department.

Awarded in August 2015

Advised by: Prof. Constantine Caramanis and Prof. Sriram Vishwanath

Thesis topic: Resource-Constrained, Scalable Learning

Technical University of Crete, Chania, Greece

MSc., ECE department.

2008 - 2010

Diploma, Electronic and Computer Engineering (5 year degree),

2002 - 2008

Advisor: Professor Nikos D. Sidiropoulos

RESEARCH GRANTS

RESEARCH GRANTS During my first seven years of tenure-track work, I was awarded a total of about 90 PhD years in competitive funding (about 2.8 million CAD after subtracting any overhead), to be disbursed over a period of 10 years.

- Google x Mila Research grant, awarded 2025-09 — \$60,000
- Meta collaborative grant, awarded 2024-03 — \$91,000
- CIFAR Canada AI chair, (originally awarded December 2018 for five years): Phase II awarded 2023-04, until 2028 — \$1,050,000
- Google x Mila Research grant, awarded 2023-09 — \$50,000
- Microsoft Research collaborative grant, awarded 2021-01 — \$54,000
- Samsung-Mila collaborative three-year grant, awarded 2020-09 — \$240,000 received (of \$2,400,000 total consortium funding)
- IVADO Postdoctoral Scholarship, for my postdoc K. Ahuja, awarded 2020-12 — \$150,000
- CIFAR Catalyst Grant, in collaboration with Manuela Girotti and Murat Erdogdu (UofT, Vector), awarded 2020-09 — \$0 received (co-applicant; \$50,000 total)
- IVADO Postdoctoral Scholarship, Fellow tier, for my postdoc N. Loizou, awarded 2020-01 — \$180,000
- Microsoft Research collaborative grant, awarded 2019-08 — \$54,000
- NSERC Discovery, awarded 2019-04 (+ competitive accelerator supplement) — \$127,500
- CIFAR Canada AI chair, awarded 2018-04 — \$1,052,500
- Fonds de Recherche du Québec, Nature et technologies, Nouveau Chercheur, awarded 2018-04 — \$50,800
- IVADO startup grant (Apogée), awarded 2017-09 — \$91,875
- Université de Montréal startup grant, awarded 2017-09 — \$70,000

DOCTORAL STUDENTS

Ryan D’Orazio (PhD candidate)
Divyat Mahajan (PhD candidate)
Zichu Liu (PhD student)
Ahmed (Mehdi) Inane (PhD student)

Mehrab Hamidi (PhD student)
Yukti Makhija (PhD student)
Mahdi Ghaznavi (PhD student, starting January ’27)

PAST STUDENTS AND SUPERVISEES

Nicolas Loizou (postdoctoral scholar, 2021-12 → assistant prof. at Johns Hopkins Univ.)

Manuela Girotti (postdoctoral scholar, 2021-07 → assistant prof. at Emory University, Atlanta, GA)

Kartik Ahuja (postdoctoral scholar, 2022-09 → research scientist at FAIR Paris)

Kilian Fatras (postdoctoral scholar, 2024-01 → research scientist at Dreamfold, Montreal)

Alexia Jolicoeur-Martineau (PhD, 2023-01 → research scientist, Microsoft Research)

Reyhane Askari Hemmat (PhD, 2024-05 → Meta, Montreal)

Adam Ibrahim (PhD, 2024-05 → Google DeepMind, Mountain View)

Charles Guille-Escuret (PhD, 2025-01 → MBZUAI Institute of Foundation Models)

Hiroki Naganuma (PhD, 2026-07 → NVIDIA)

Vincent Quirion (MSc, 2024-2026 → Soma Energy)

Mehrnaz Mofakhami (MSc, 2024-08 → research assistant at Mila)

Rémi Piché-Taillefer (MSc, 2021-06 → Microsoft Research, Montréal)

Brady Neal (MSc, 2019-12 → Senior Research Scientist at Dataiku)

Oscar Clivio (research intern, 2024-11 → PhD student, University of Oxford)

Lyle Kim (research intern, 2023-01 → PhD candidate, Rice University)

Vincent Quirion (research intern, 2024-08 → MSc at UdeM/Mila, in progress)

Ange-Clement Akazan (research intern, 2023-12 → MSc student at the African Institute of Mathematical Sciences)

Séb Arnold (research intern, 2018-08 → PhD candidate at USC)

Vinayak Tantia (research intern, 2018-08 → research engineer, FAIR Montréal)

Baptiste Goujaud (research intern, 2020-03 → PhD candidate, École Polytechnique)

Amartya Mitra (research intern, 2020-06 → ML Researcher at Morgan Stanley)

Nicolas Gagné (research intern, 2018-08 → PhD candidate at McGill)

TEACHING

University of Montreal

Latest teaching evaluation: 3.7/4.0

Fundamentals of machine learning (in french) **Fall 2026**

Fundamentals of machine learning (in french) **Fall 2025**

Fundamentals of machine learning (in french) **Fall 2024**

Fundamentals of machine learning **Fall 2023**

Theoretical principles for deep learning **Winter 2023**

Fundamentals of machine learning **Fall 2022**

Theoretical principles for deep learning **Winter 2022**

Fundamentals of machine learning **Fall 2021**

Theoretical principles for deep learning **Winter 2021**

Fundamentals of machine learning **Fall 2020**

<i>Theoretical principles for deep learning</i>	Winter 2020
<i>Fundamentals of machine learning</i>	Fall 2019
<i>Theoretical principles for deep learning</i>	Winter 2019
<i>Fundamentals of machine learning</i>	Fall 2018
<i>Theoretical principles for deep learning</i>	Winter 2018
The University of Texas at Austin	
<i>Teaching Assistant—Information Theory</i>	Spring 2012
Technical University of Crete	
<i>Teaching Assistant—Telecommunication Networks</i>	Fall 2008

PUBLICATIONS

- D. Mahajan, S. Goyal, B. Youbi Idrissi, M. Pezeshki, **I. Mitliagkas**, D. Lopez-Paz, K. Ahuja
Beyond Multi-Token Prediction: Pretraining LLMs with Future Summaries
arXiv:2510.14751
International Conference on Learning Representations (ICLR), 2026
- N. Tsikouras, Y. Pantis, **I. Mitliagkas**, C. Tzamos
A Derandomization Framework for Structure Discovery: Applications in Neural Networks and Beyond
arXiv:2510.19382
International Conference on Learning Representations (ICLR), 2026
- K. Yoshida, Y. Naraki, T. Horie, R. Shimizu, **I. Mitliagkas**, H. Naganuma
DisTaC: Conditioning Task Vectors via Distillation for Robust Model Merging
International Conference on Learning Representations (ICLR), 2026
- A.M. Inane, V. Quirion, G.K. Dziugaite, **I. Mitliagkas**
Unlearning with Asymmetric Sources: Improved Unlearning-Utility Trade-off with Public Data
arXiv:2605.11170
International Conference on Machine Learning (ICML), 2026
- H. Naganuma, S. Gupta, Y. Briki, **I. Mitliagkas**, I. Rish, P. Raman, H.J.M. Shi
Adaptive Batch Sizes Using Non-Euclidean Gradient Noise Scales for Stochastic Sign and Spectral Descent
arXiv:2602.03001
International Conference on Machine Learning (ICML), 2026
- H. Naganuma, R. Hataya, K. Yoshida, **I. Mitliagkas**
An Empirical Study of Pre-trained Model Selection for Out-of-Distribution Generalization and Calibration
arXiv:2307.08187
Transactions on Machine Learning Research (TMLR), 2025
- L. Maes, TH. Zhang, A. Jolicoeur-Martineau, **I. Mitliagkas**, D. Scieur
Understanding Adam Requires Better Rotation Dependent Assumptions
arXiv:2410.19964
Neural Information Processing Systems (NeurIPS), 2025

- H. Naganuma, X. Zhang, MC. Yue, **I. Mitliagkas**, PA. Witte, RJ. Hewett
Pseudo-Asynchronous Local SGD: Robust and Efficient Data-Parallel Training
arXiv:2504.18454
Transactions on Machine Learning Research (TMLR), 2025
- D. Mahajan, M. Pezeshki, C. Arnal, **I. Mitliagkas**, K. Ahuja, P. Vincent
Compositional risk minimization
arXiv:2410.06303
International Conference on Machine Learning (ICML), 2025
- D. Beaglehole, **I. Mitliagkas**, A. Agarwala
Feature learning as alignment: a structural property of gradient descent in non-linear neural networks
arXiv:2402.05271
Transactions on Machine Learning Research (TMLR), 2024
- E. Triantafillou, P. Kairouz, F. Pedregosa, J. Hayes, M. Kurmanji, K. Zhao, V. Dumoulin, J.J. Junior, **I. Mitliagkas**, J. Wan, L.S. Hosoya, S. Escalera, G.K. Dziugaite, P. Triantafillou, I. Guyon
Are we making progress in unlearning? Findings from the first NeurIPS unlearning competition
arXiv:2406.09073
Neural Information Processing Systems (NeurIPS), 2024
- R. D’Orazio, D. Vucetic, Z. Liu, JL. Kim, **I. Mitliagkas**, G. Gidel
Solving hidden monotone variational inequalities with surrogate losses
arXiv:2411.05228
International Conference on Learning Representations (ICLR), 2025
- C. Guille-Escuret, PA. Noël, **I. Mitliagkas**, D. Vazquez, J. Monteiro
Expecting The Unexpected: Towards Broad Out-Of-Distribution Detection
arXiv:2308.11480
Neural Information Processing Systems (NeurIPS), 2024
- C. Guille-Escuret, H. Naganuma, K. Fatras, **I. Mitliagkas**
No Wrong Turns: The Simple Geometry Of Neural Networks Optimization Paths
arXiv:2306.11922
International Conference on Machine Learning (ICML), 2024
- D. Mahajan, **I. Mitliagkas**, B. Neal, V. Syrgkanis
Empirical Analysis of Model Selection for Heterogeneous Causal Effect Estimation
arXiv:2210.01210
International Conference on Learning Representations (ICLR), 2024 [**spotlight presentation**]
- R. Askari Hemmat*, A. Mitra*, G. Lajoie, **I. Mitliagkas**
LEAD: Least-Action Dynamics for Min-Max Optimization
arXiv:2010.13846
International Conference on Learning Representations (ICLR), 2024 (invited); *Transactions on Machine Learning Research (TMLR)* [featured]
- S. Lachapelle, D. Mahajan, **I. Mitliagkas**, S. Lacoste-Julien
Additive Decoders for Latent Variables Identification and Cartesian-Product Extrapolation
Neural Information Processing Systems (NeurIPS), 2023 [**oral presentation**]

- C.G. Escuret, P. Rodriguez, D. Vazquez, **I. Mitliagkas**, J. Monteiro
 CADet: Fully Self-Supervised Anomaly Detection With Contrastive Learning
 arXiv:2210.01742
Neural Information Processing Systems (NeurIPS), 2023
- T. Salvador, K. Fatras, **I. Mitliagkas**, A. Oberman
 A Reproducible and Realistic Evaluation of Partial Domain Adaptation Methods
 arXiv:2210.01210
Transactions on Machine Learning Research (TMLR), 2023
- R. D’Orazio, N. Loizou, I. Laradji, **I. Mitliagkas**
 Stochastic Mirror Descent: Convergence Analysis and Adaptive Variants via the Mirror
 Stochastic Polyak Stepsize
 arXiv:2110.15412
Transactions on Machine Learning Research (TMLR), 2023
- S. Lachapelle, T. Deleu, D. Mahajan, **I. Mitliagkas**, Y. Bengio, S. Lacoste-Julien,
 Q. Bertrand
 Synergies Between Disentanglement and Sparsity: a Multi-Task Learning Perspective
 arXiv:2304.06879
International Conference on Machine Learning (ICML), 2023
- H. Naganuma, K. Ahuja, S. Takagi, T. Motokawa, R. Yokota, K. Ishikawa, I. Sato,
I. Mitliagkas
 Empirical Study on Optimizer Selection for Out-of-Distribution Generalization
 arXiv:2211.08583
Transactions on Machine Learning Research (TMLR), 2023
- S. Sokota, R. D’Orazio, J. Z. Kolter, N. Loizou, M. Lanctot, **I. Mitliagkas**, N. Brown,
 C. Kroer
 A Unified Approach to Reinforcement Learning, Quantal Response Equilibria, and Two-
 Player Zero-Sum Games
 arXiv:2206.05825
International Conference on Learning Representations (ICLR), 2023
- A. Mousavi-Hosseini, S. Park, M. Girotti, **I. Mitliagkas**, M. A. Erdogdu
 Neural Networks Efficiently Learn Low-Dimensional Representations with SGD
International Conference on Learning Representations (ICLR), 2023
- M. Mofakhami, **I. Mitliagkas**, G. Gidel
 Performative Prediction with Neural Networks
Artificial Intelligence and Statistics (AISTATS), 2023
- C. Guille-Escuret, A. Ibrahim, B. Goujaud, **I. Mitliagkas**
 Gradient Descent Is Optimal Under Lower Restricted Secant Inequality And Upper Er-
 ror Bound
Neural Information Processing Systems (NeurIPS), 2022
- K. Fatras, H. Naganuma, **I. Mitliagkas**
 Optimal transport meets noisy label robust loss and MixUp regularization for domain
 adaptation
 arXiv:2206.11180
Conference on Lifelong Learning Agents (CoLLAs), 2022

- K. Ahuja, D. Mahajan, V. Syrgkanis, **I. Mitliagkas**
Towards efficient representation identification in supervised learning
Conference on Causal Learning and Reasoning (CLear), 2022
- N. Loizou, H. Berard, G. Gidel, **I. Mitliagkas**, S. Lacoste-Julien
Stochastic Gradient Descent-Ascent and Consensus Optimization for Smooth Games:
Convergence Analysis under Expected Co-coercivity
arXiv:2107.00052
Neural Information Processing Systems (NeurIPS), 2021
- K. Ahuja, E. Caballero, D. Zhang, Y. Bengio, **I. Mitliagkas**, I. Rish
Invariance Principle Meets Information Bottleneck for Out-of-Distribution Generalization
arXiv:2106.06607
Neural Information Processing Systems (NeurIPS), 2021 [**spotlight presentation**]
- A. Jolicoeur-Martineau, R. Piche-Taillefer, **I. Mitliagkas**, R. Tachet des Combes
Adversarial score matching and improved sampling for image generation
arXiv:2009.05475
International Conference on Learning Representations (ICLR), 2021
- C. Guille-Escuret*, B. Goujaud*, M. Girotti, **I. Mitliagkas**
A Study of Condition Numbers for First-Order Optimization
arXiv:2012.05782
Artificial Intelligence and Statistics (AISTATS), 2021 [best student paper award OPT2020]
- G. K. Dziugaite, A. Drouin, B. Neal, N. Rajkumar, E. Caballero, L. Wang, **I. Mitliagkas**,
D. Roy
In search of robust measures of generalization
arXiv:2010.11924
Neural Information Processing Systems (NeurIPS), 2020
- N. Loizou, H. Berard, A. Jolicoeur-Martineau, P. Vincent, S. Lacoste-Julien, **I. Mitliagkas**
Stochastic Hamiltonian Gradient Methods for Smooth Games
International Conference on Machine Learning (ICML), 2020
- A. Ibrahim, W. Azizian, G. Gidel, **I. Mitliagkas**
Linear Lower Bounds and Conditioning of Differentiable Games
arXiv:1906.07300
International Conference on Machine Learning (ICML), 2020
- W. Azizian, D. Scieur, **I. Mitliagkas**, S. Lacoste-Julien, G. Gidel
Accelerating Smooth Games by Manipulating Spectral Shapes
arXiv:2001.00602
Artificial Intelligence and Statistics (AISTATS), 2020
- W. Azizian, **I. Mitliagkas**, S. Lacoste-Julien, G. Gidel
A Tight and Unified Analysis of Gradient-Based Methods for a Whole Spectrum of
Differentiable Games
arXiv:1906.05945
Artificial Intelligence and Statistics (AISTATS), 2020
- S. M. Arnold, P. A. Manzagol, R. Babanezhad, **I. Mitliagkas**, N. L. Roux
Reducing the variance in online optimization by transporting past gradients
arXiv:1901.08680
Neural Information Processing Systems (NeurIPS), 2019 [**spotlight presentation**]

- I. Albuquerque, J. Monteiro, T. Doan, B. Considine, T. Falk, **I. Mitliagkas**
 Multi-objective training of Generative Adversarial Networks
 arXiv:1901.08680
International Conference on Machine Learning (ICML), 2019
- V. Verma, A. Lamb, C. Beckham, A. Najafi, **I. Mitliagkas**, A. Courville, D. Lopez-Paz, Y. Bengio
 Manifold Mixup: Better Representations by Interpolating Hidden States
 arXiv:1707.02392
International Conference on Machine Learning (ICML), 2019
- A. Lamb, J. Binas, A. Goyal, S. Subramanian, **I. Mitliagkas**, Y. Bengio, M. Mozer
 State-Reification Networks: Improving Generalization by Modeling the Distribution of Hidden Representations
 arXiv:1905.11382
International Conference on Machine Learning (ICML), 2019 [oral presentation]
- B. Kanuparthi, D. Arpit, G. Kerg, N. R. Ke, **I. Mitliagkas**, Y. Bengio
 h-detach: Modifying the LSTM Gradient Towards Better Optimization
International Conference on Learning Representations (ICLR), 2019
- G. Gidel, R. Askari, M. Pezeshki, G. Huang, S. Lacoste-Julien, **I. Mitliagkas**
 Negative Momentum for Improved Game Dynamics
 arXiv:1807.04740
Artificial Intelligence and Statistics (AISTATS), 2019
- J. Zhang, **I. Mitliagkas**
 YellowFin and the Art of Momentum Tuning
Systems and Machine Learning (SysML), 2019
- P. Achlioptas, O. Diamanti, **I. Mitliagkas**, L. Guibas
 Learning Representations and Generative Models for 3D Point Clouds
 arXiv:1707.02670
International Conference on Machine Learning (ICML), 2018
- J. Zhang, **I. Mitliagkas**
 YellowFin: Adaptive optimization for (A)synchronous systems
 arXiv:1706.03471
Systems and Machine Learning (SysML), 2018 [oral presentation]
- C. De Sa, B. He, **I. Mitliagkas**, C. Ré, P. Xu
 Accelerated stochastic power iteration
 arXiv:1707.02670
Artificial Intelligence and Statistics (AISTATS), 2018
- T. Kurth, J. Zhang, N. Satish, **I. Mitliagkas**, E. Racah, M.A. Patwary, T. Malas, N. Sundaram, W. Bhimji, M. Smorkalov, J. Deslippe, M. Shiryaev, S. Sridharan, P. Dubey
 Deep Learning at 15PF: Supervised and Semi-Supervised Classification for Scientific Data
 arXiv:1708.05256
Supercomputing (SC), 2017

- I. Mitliagkas**, L. Mackey
Improving Gibbs Sampler Scan Quality with DoGS
arXiv:1707.05807
International Conference on Machine Learning (ICML), 2017
- I. Mitliagkas**, C. Zhang, S. Hadjis, C. Ré
Asynchrony begets Momentum, with an Application to Deep Learning
arXiv:1605.09774
Allerton Conference on Communication, Control, and Computing (Allerton), 2016
- B. He, C. De Sa, **I. Mitliagkas**, C. Ré
Scan Order in Gibbs Sampling: Models in Which it Matters and Bounds on How Much
arXiv:1606.03432
Neural Information Processing Systems (NeurIPS), 2016
- I. Mitliagkas**, M. Borokhovich, A. Dimakis, C. Caramanis
FrogWild! – Fast PageRank Approximations on Graph Engines
International Conference on Very Large Data Bases (VLDB), 2015
- D. Papailiopoulos, **I. Mitliagkas**, A. Dimakis, C. Caramanis
Finding dense subgraphs through low-rank approximations
International Conference on Machine Learning (ICML), 2014
- I. Mitliagkas**, C. Caramanis, P. Jain
Memory-limited Streaming PCA
Neural Information Processing Systems (NeurIPS), 2013
- I. Mitliagkas**, A. Gopalan, C. Caramanis, S. Vishwanath
User Rankings from Comparisons: Learning Permutations in High Dimensions
Allerton Conference on Communication, Control, and Computing (Allerton), 2011
- I. Mitliagkas**, N. D. Sidiropoulos, and A. Swami
Joint Power and Admission Control for Ad-hoc and Cognitive Underlay Networks: Convex Approximation and Distributed Implementation
IEEE Transactions on Wireless Communications (TWC), 2011
- I. Mitliagkas**, S. Vishwanath
Strong Information-Theoretic Limits for Source/Model Recovery
Allerton Conference on Communication, Control, and Computing (Allerton), 2010
- I. Mitliagkas**, N. D. Sidiropoulos, and A. Swami
Distributed Joint Power and Admission Control for Ad-hoc and Cognitive Underlay Networks
International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2010
- I. Mitliagkas**, N. D. Sidiropoulos, and A. Swami
Convex Approximation-based Joint Power and Admission Control for Cognitive Underlay Networks
International Wireless Communications and Mobile Computing Conference (IWCMC), 2008

T. Nakamori, L. Gomezjurado Gonzalez, G. Talluri, A. Tiwari, H. Kawashima, **I. Mitliagkas**,
G. Rabusseau, H. Naganuma
Orth-Dion: Eliminating Geometric Mismatch in Distributed Low-Rank Spectral Optimization
ICML 2026 CoLoRAI Workshop, arXiv preprint arXiv:2605.16341

H. Naganuma, J.L. Kim, A. Kyrillidis, **I. Mitliagkas**
Smoothness-adaptive sharpness-aware minimization for finding flatter minima
5th Workshop on practical ML for limited/low resource settings

M. Mofakhami, R. Bayat, **I. Mitliagkas**, J. Monteiro, V. Zantedeschi
Performance Control in Early Exiting to Deploy Large Models at the Same Cost of Smaller Ones
ICML 2024 workshop on Efficient Systems for Foundational Models, arXiv preprint arXiv:2412.19325

B. Neal, **I. Mitliagkas**
In Support of Over-Parametrization in Deep Reinforcement Learning: an Empirical Study
ICML 2019 Workshop on Identifying and Understanding Deep Learning Phenomena

B. Neal, S. Mittal, A. Baratin, V. Tania, M. Scicluna, S. Lacoste-Julien, **I. Mitliagkas**
A Modern Take on the Bias-Variance Tradeoff in Neural Networks
ICML 2019 Workshop on Identifying and Understanding Deep Learning Phenomena, arXiv preprint arXiv:1810.08591

J. Zhang, C. De Sa, **I. Mitliagkas**, C. Ré
Parallel SGD: When does averaging help?
Optimization Methods for the Next Generation of Machine Learning Workshop, ICML 2016, arXiv preprint arXiv:1606.07365

Z. Liu, N. Okolo, R. D’Orazio, D. Vucetic, **I. Mitliagkas**, G. Gidel
A Surrogate Perspective on Convergence of Fixed-Target DQN
Submitted to NeurIPS 2026 (under review)

H. Naganuma, L. Gomezjurado Gonzalez, M. Ghaznavi, A. Nitanda, **I. Mitliagkas**,
S.P. Liew, R. Hataya
Which Geometry on Which Layer? A Principled Criterion for Mixed-Optimizer Training
Submitted to NeurIPS 2026 (under review)

S. Dufort-Labbé, M. Hamidi, R. Pascanu, **I. Mitliagkas**, D. Scieur, A. Baratin
Navigating Potholes with Geometry-Aware Sharpness Minimization
arXiv preprint arXiv:2605.16134

S. Nakai, Y. Fadhloun, K. Mathlouthi, K. Yoshida, G. Talluri, **I. Mitliagkas**, H. Naganuma
Revisiting Generalization Measures Beyond IID: An Empirical Study under Distributional Shift
arXiv preprint arXiv:2602.01718

O. Clivio, D. Mahajan, P. Taslakian, S. Magliacane, **I. Mitliagkas**
Learning to Defer for Causal Discovery with Imperfect Experts
arXiv preprint arXiv:2502.13132

AC. Akazan, A. Jolicoeur-Martineau, **I. Mitliagkas**
Generating Tabular Data Using Heterogeneous Sequential Feature Forest Flow Matching
arXiv preprint arXiv:2410.15516

A. Ibrahim, C. Guille-Escuret, **I. Mitliagkas**, I. Rish, D. Krueger, P. Bashivan
Towards Out-of-Distribution Adversarial Robustness
arXiv preprint arXiv:2210.03150

M. Girotti, **I. Mitliagkas**, Gauthier Gidel
Convergence Analysis and Implicit Regularization of Feedback Alignment for Deep Linear Networks
arXiv preprint arXiv:2110.10815

A. Jolicoeur-Martineau, K. Li*, R. Piché-Taillefer*, T. Kachman*, **I. Mitliagkas**
Gotta Go Fast When Generating Data with Score-Based Models
arXiv preprint arXiv:2105.14080

I. Albuquerque, J. Monteiro, T. Falk, **I. Mitliagkas**
Generalizing to unseen domains via distribution matching
arXiv preprint arXiv:1911.00804

A. Jolicoeur-Martineau, **I. Mitliagkas**
Gradient penalty from a maximum margin perspective
arXiv preprint arXiv:1910.06922

S. Hadjis, C. Zhang, **I. Mitliagkas**, C. Ré
Omnivore: An Optimizer for Multi-device Deep Learning on CPUs and GPUs
Technical report, arXiv:1606.04487

Trudeau meets with newly appointed Canada CIFAR AI Chairs, **CIFAR News**

Sept chercheurs de l'UdeM titulaires de chaires en intelligence artificielle Canada-CIFAR, **UdeM Nouvelles**

NERSC Scales Scientific Deep Learning to 15 Petaflops, **HPC Wire**

De la Grèce à l'UdeM: l'étonnant parcours d'Ioannis Mitliagkas, **UdeM Nouvelles**

AWARDS, DISTINCTIONS

Recognized among top reviewers of ICML, 2026

NeurIPS Foundation, top 10% of reviewers, 2020

CIFAR Canada AI chair

NIPS Foundation, listed among best reviewers, 2018

Gerondelis Foundation Inc.: Graduate Scholarship, 2014

The University of Texas at Austin: Microelectronics and Computer Development (MCD) Fellowship, 2009-2011

Technical University of Crete: Undergraduate excellence award, 2008

State Scholarships Foundation (Greece): Undergraduate excellence award, 2005

Technical Chamber of Greece: Undergraduate excellence award, 2005

PROFESSIONAL SERVICE

Archimedes summer school, Athens, Greece (2022)

Neuromatch Academy Deep Learning Track, World-wide (2021, 2022)

AI4Good Summer School, Canada (2021)

Co-organizer of INFORMS 2021 workshop:
“Accelerated methods in convex and non-convex optimization”

Co-founder and organizer of **MTL MLOpt**, a Montreal-based group of researchers from many universities and industrial labs.

Member of the inaugural program committee of MLSys:
The committee's role was to decide the conference's focus and steer its future goals.

Organizer of NeurIPS 2018, 2019 workshop:
“Smooth Games Optimization and Machine Learning”

Reviewer of MITACS Accelerate grants

Served as head of the scientific committee in charge of evaluating IVADO grants (2018) and as member in subsequent years.

Host of oral session, NeurIPS 2021

Reviewer and AC for a number of journals and conferences including NeurIPS, ICML, COLT, AISTATS, ALT, AAAI, ICLR, JMLR (editorial board), IJCAI, SIGGRAPH, Transactions on Information Theory, ISIT, ICASSP, Transactions on Wireless Communications.

RECENT INVITED
TALKS (NOT
INCLUDING
ACCEPTED PAPER
PRESENTATIONS)

Neuromatch Academy Deep Learning Track, World-wide	Summer 2023
Archimedes summer school, Athens, Greece	July 2022
Simons semester in game theory, Berkeley, CA	January-March, 2022
SIAM conference in optimization OP21, Spokane, WA	July 2021
Neuromatch Academy Deep Learning Track, World-wide	Summer 2021
AI4Good Summer School, Canada	May 2021
Google Brain, Montreal, QC	November 2020
ITA, San Diego, CA	February 2020
INFORMS, Seattle, WA	October 2019
Microsoft Research workshop, Montréal, QC	October 2019
Theoretical Advances in Deep Learning, Workshop, Istanbul	July 2019
UT Austin, TX	March 2019
NVIDIA, Webinar	March 2019
ElementAI, Toronto, ON	November 2018
BorealisAI, Toronto, ON	October 2018
USC, Los Angeles, CA	October 2018
Microsoft Research workshop, Montréal, QC	October 2018
ElementAI, Toronto, ON	September 2018
Microsoft Research, Montréal, QC	August 2018
ElementAI, Montréal, QC	June 2018
FAIR, Montréal, QC	May 2018
RLLab, McGill, Montréal, QC	April 2018
ElementAI, Montréal, QC	April 2018
TechAide, Montréal, QC	April 2018
ECE Seminar, UT Austin, TX	March 2018
BayesComp, Barcelona Spain	March 2018
SysML, Stanford CA	February 2018
Google Brain, Montréal	November 2017
Texas Wireless Summit, Austin, TX	October 2017
Colloquium, University of Montréal	September 2017
Colloquium, The University of Texas, Austin	September 2017
AutoML workshop, ICML, Sydney	August 2017
Workshop on Advances in Computing Architectures, Stanford SystemX	April 2016
ITA workshop, San Diego, CA	February 2017
AAAI 2017 Workshop on Distributed Machine Learning	February 2017
Microsoft Research, Cambridge, UK	December 2016
SystemX Stanford Alliance Fall Conference	November 2016
Microsoft Research, New England	October 2016
Allerton Conference, Monticello, IL	September 2016
Google Brain, Mountain View, CA	August 2016
MIT Lincoln Labs, MA	August 2016
NVIDIA, Santa Clara, CA	July 2016

